AWS High Availability & Disaster Recovery — Important Interview Questions & Answers

1.What is High Availability?
High Availability means designing an application so that it remains available even when some components fail.
It is usually achieved by using multiple Availability Zones and redundant resources.
Simple ga gurthupettuko:
High Availability ante oka resource fail ayina application available ga continue avvadam.
Eng: High Availability keeps an application available even when some components fail.

2.What is Disaster Recovery?
Disaster Recovery is the process of restoring applications and data after a major failure or disaster.
It focuses on recovering the workload and bringing the application back to normal operation.
Simple ga gurthupettuko:
Disaster Recovery ante major failure tarvata application and data ni recover cheyyadam.
Eng: Disaster Recovery is the process of recovering applications and data after a disaster.

3.What is Multi-AZ?
Multi-AZ means deploying resources across multiple Availability Zones within an AWS Region.
It helps improve availability and provides protection against an Availability Zone failure.
Simple ga gurthupettuko:
Multi-AZ ante multiple Availability Zones lo resources deploy chesi availability improve cheyyadam.
Eng: Multi-AZ deployment improves availability by using multiple Availability Zones.

4.What is Multi-Region?
Multi-Region means deploying an application or its components across multiple AWS Regions.
It can provide protection against a Region-level failure and can also reduce latency for global users.
Simple ga gurthupettuko:
Multi-Region ante application ni multiple AWS Regions lo deploy cheyyadam.
Eng: Multi-Region deployment protects against Region-level failures and supports global applications.

5.What is RTO?
RTO stands for Recovery Time Objective.
It defines the maximum acceptable time required to restore an application after a failure.
Simple ga gurthupettuko:
RTO ante failure tarvata application entha time lo recover avvali ane target.
Eng: RTO defines the maximum acceptable time to restore an application.

6.What is RPO?
RPO stands for Recovery Point Objective.
It defines the maximum acceptable amount of data loss measured in time after a failure.
Simple ga gurthupettuko:
RPO ante maximum entha old data varaku loss accept cheyyagalamo ane target.
Eng: RPO defines the maximum acceptable amount of data loss after a failure.

7.What is the difference between RTO and RPO?
RTO defines how quickly the application must be recovered.
RPO defines how much data loss is acceptable.
Simple ga gurthupettuko:
RTO ante recovery time, RPO ante data loss time.
Eng: RTO is about recovery time, while RPO is about acceptable data loss.

8.What is Failover?
Failover is the process of switching traffic or workload from a failed primary resource to a healthy secondary resource.
It helps maintain application availability during failures.
Simple ga gurthupettuko:
Failover ante primary fail ayite traffic ni healthy secondary resource ki shift cheyyadam.
Eng: Failover switches traffic from a failed resource to a healthy secondary resource.

9.What is Backup?
A Backup is a copy of data created so that it can be restored after accidental deletion, corruption, or failure.
AWS provides different backup options depending on the service and workload.
Simple ga gurthupettuko:
Backup ante future lo restore cheyyadaniki data yokka copy maintain cheyyadam.
Eng: A Backup is a copy of data used for recovery.

10.What is the difference between Backup and Disaster Recovery?
Backup focuses mainly on protecting and restoring data.
Disaster Recovery focuses on restoring the complete application or workload, including infrastructure, data, and required services.
Simple ga gurthupettuko:

Backup data recovery kosam, Disaster Recovery complete application recovery kosam.
Eng: Backup protects data, while Disaster Recovery restores the overall workload.

11.What is the difference between High Availability and Disaster Recovery?
High Availability focuses on keeping an application running during component or Availability Zone failures.
Disaster Recovery focuses on recovering the application after larger failures such as major infrastructure or Region-level disasters.
Simple ga gurthupettuko:
High Availability continuous availability kosam, Disaster Recovery major failure tarvata recovery kosam.
Eng: High Availability maintains service availability, while Disaster Recovery focuses on recovery after major failures.

12.What are common AWS Disaster Recovery strategies?
Common approaches include Backup and Restore, Pilot Light, Warm Standby, and Multi-Site Active/Active.
They provide different levels of recovery speed, complexity, and cost.
Simple ga gurthupettuko:
DR strategies ante Backup & Restore, Pilot Light, Warm Standby, Multi-Site Active/Active.
Eng: Common AWS DR strategies include Backup and Restore, Pilot Light, Warm Standby, and Multi-Site Active/Active.

13.What is Backup and Restore?
Backup and Restore means keeping backups of the workload and restoring the required resources when a disaster occurs.
It is generally simpler and lower cost but usually has a longer recovery time.
Simple ga gurthupettuko:
Backup and Restore ante backups maintain chesi disaster vachinappudu resources restore cheyyadam.
Eng: Backup and Restore uses backups to recover resources after a disaster.

14.What is Pilot Light?
Pilot Light keeps the core components of an application ready while most resources are created or scaled up during recovery.
It provides faster recovery than a basic Backup and Restore approach.
Simple ga gurthupettuko:
Pilot Light ante important core components ready ga petti, disaster time lo remaining resources start cheyyadam.
Eng: Pilot Light keeps core components ready for faster disaster recovery.

15.What is Warm Standby?
Warm Standby keeps a scaled-down version of the production environment running in another location.
During a failure, it can be scaled up to handle production traffic.
Simple ga gurthupettuko:
Warm Standby ante smaller production environment already running lo petti failure time lo scale up cheyyadam.
Eng: Warm Standby keeps a smaller running environment ready for recovery.

16.What is Multi-Site Active/Active?
Multi-Site Active/Active means running workloads actively in multiple locations at the same time.
Traffic can be distributed across the locations, providing very fast recovery when one location fails.
Simple ga gurthupettuko:
Active/Active ante multiple locations simultaneously traffic handle cheyyadam.
Eng: Multi-Site Active/Active runs workloads in multiple locations simultaneously.

17.How would you design a highly available AWS application?
I would deploy application resources across multiple Availability Zones and use a Load Balancer to distribute traffic.
I would also use Auto Scaling, health checks, backups, monitoring, and appropriate database high-availability features.
Simple ga gurthupettuko:
HA kosam Multi-AZ + Load Balancer + Auto Scaling + Health Checks + Monitoring use chestham.
Eng: I would use Multi-AZ, load balancing, Auto Scaling, health checks, backups, and monitoring.

18.How would you design disaster recovery for a critical application?
I would first define the required RTO and RPO based on business requirements.
Then I would choose an appropriate DR strategy, maintain backups or a secondary environment, and regularly test the recovery process.
Simple ga gurthupettuko:
Critical application ki first RTO/RPO decide chesi, suitable DR strategy and regular recovery testing implement chestham.
Eng: I would define RTO and RPO, choose a suitable DR strategy, and regularly test recovery.

19.What is the most important thing to remember about High Availability and Disaster Recovery?
High Availability keeps applications running during failures, while Disaster Recovery focuses on recovering applications after major failures.
RTO, RPO, Multi-AZ, Multi-Region, backups, and failover are important concepts.
Simple ga gurthupettuko:
HA ante application available ga maintain cheyyadam, DR ante failure tarvata application ni recover cheyyadam.
Eng: High Availability keeps applications running, while Disaster Recovery restores them after major failures.